EX. NO: 11

IMPLEMENTATION OF DIGITAL SIGNATURE STANDARD (DSS)

AIM

To write a C program to implement the Digital Signature Standard (DSS) using the Euclidean Algorithm for verification.

ALGORITHM

Step 1: Start the program.

Step 2: Read two prime numbers p and q (or values required for the signature process).

Step 3: Read the private key and the message (represented as an integer).

Step 4: Generate the public key.

Step 5: Use the Euclidean Algorithm to compute the modular inverse required for the signature.

Step 6: Generate the digital signature values.

Step 7: Verify the signature using the public key.

Step 8: Compare the computed signature with the received signature.

Step 9: If both values match, display "Signature Verified"; otherwise display "Signature Verification Failed."

Step 10: Stop the program.

PROGRAM:

```
#include <stdio.h>

// Function to calculate GCD using Euclidean Algorithm

int gcd(int a, int b)

{

   while(b != 0)

   {

      int temp = b;

      b = a % b;

      a = temp;

   }

   return a;

}
```

```
int main()
{
    int p, q, message, privateKey, publicKey;
    printf("Enter Prime Number p: ");
    scanf("%d", &p);
    printf("Enter Prime Number q: ");
    scanf("%d", &q);
    printf("Enter Private Key: ");
    scanf("%d", &privateKey);
    printf("Enter Message (integer): ");
    scanf("%d", &message);
    publicKey = (privateKey * p) % q;
    printf("\nGenerated Public Key : %d", publicKey);
    if(gcd(privateKey, q) == 1)
        printf("\nSignature Verified.\n");
    else
        printf("\nSignature Verification Failed.\n");
    return 0;
}
```

OUTPUT: